

Integration — Lesson 38 Test: Growth, Decay & Logistic

10 questions · show your work in the box · calculators allowed · exact forms with $\ln 2$ are fine · every answer needs UNITS

Name: _____ Date: _____ Score: _____ / 10

- 1.** A colony of yeast cells, N , grows at a rate proportional to the number of cells present. 1 pt
Write the differential equation, and write its solution in terms of the starting amount N_0 .

- 2.** A loaf of bread at temperature T sits in a 15° kitchen and cools at a rate proportional to how much hotter it is than the kitchen. Write the differential equation, and explain in one sentence why it needs a minus sign. 1 pt

- 3.** A patch of lily pads starts with 250 pads and doubles every 5 days. (a) Write a formula for $P(t)$ using base 2, with t in days. (b) How many pads are there after 20 days? 1 pt

4. A medicine leaves the bloodstream with a half-life of 8 hours. A patient is given a 640 mg dose. How much of the dose remains after 24 hours? 1 pt

5. A bacteria culture grows exponentially from 400 cells to 3200 cells in 12 hours. Find the doubling time, then check your answer by substituting $t = 12$ back into your model. 1 pt

6. A colony follows $P = 60 \cdot 2^{(t/4)}$ with t in hours. At what time does the population reach 1920 cells? 1 pt

7. (a) Rewrite $N = 250 \cdot 2^{(t/5)}$ in the form $N = 250 \cdot e^{(kt)}$: give k exactly, then as a decimal using $\ln 2 \approx 0.693$. (b) A sample decays according to $y = 80 \cdot e^{-(\ln 2/7)t}$ with t in days. What is its half-life, and how much remains after 21 days? 1 pt

8. A roast comes out of the oven at 180° and is left in a 20° kitchen. After 12 minutes it has cooled to 100° . (a) Find the half-life of the GAP between the roast and the kitchen. (b) Predict the roast's temperature after 36 minutes. (c) Will the roast ever be exactly 20° ? Explain using the formula $T = 20 + 160e^{(-kt)}$. 1 pt

9. A rabbit population satisfies $dP/dt = 0.3P(1 - P/900)$ with $P(0) = 150$, t in years. (a) What is the carrying capacity? (b) What is the limit of P as $t \rightarrow \infty$? (c) At what population is the herd growing fastest, and what is dP/dt there? (d) What is dP/dt when $P = 300$? 1 pt

10. A valley can support at most 4000 elk, and the herd grows logistically with $k = 0.25$ per year. (a) Write the differential equation. (b) At what herd size does the herd grow fastest, and what is that fastest growth rate? (c) A student says: "The herd grows fastest when it reaches 4000, because that is when there are the most elk." Find the error. (d) If 5000 elk were released into the valley, what does the model predict in the long run? 1 pt
